

6 Warfarin is an anticoagulant.

The dose of warfarin that is given to a patient has to be monitored regularly. This is done by adding thromboplastin to a blood sample and measuring the time it takes for the blood to clot.

The result is compared with a control sample and is reported as an International Normalised Ratio (INR) level.

(a) How many of the following statements are correct for thromboplastin?

- it is stored in platelets
- it is an active enzyme
- it is converted into thrombin

(1)

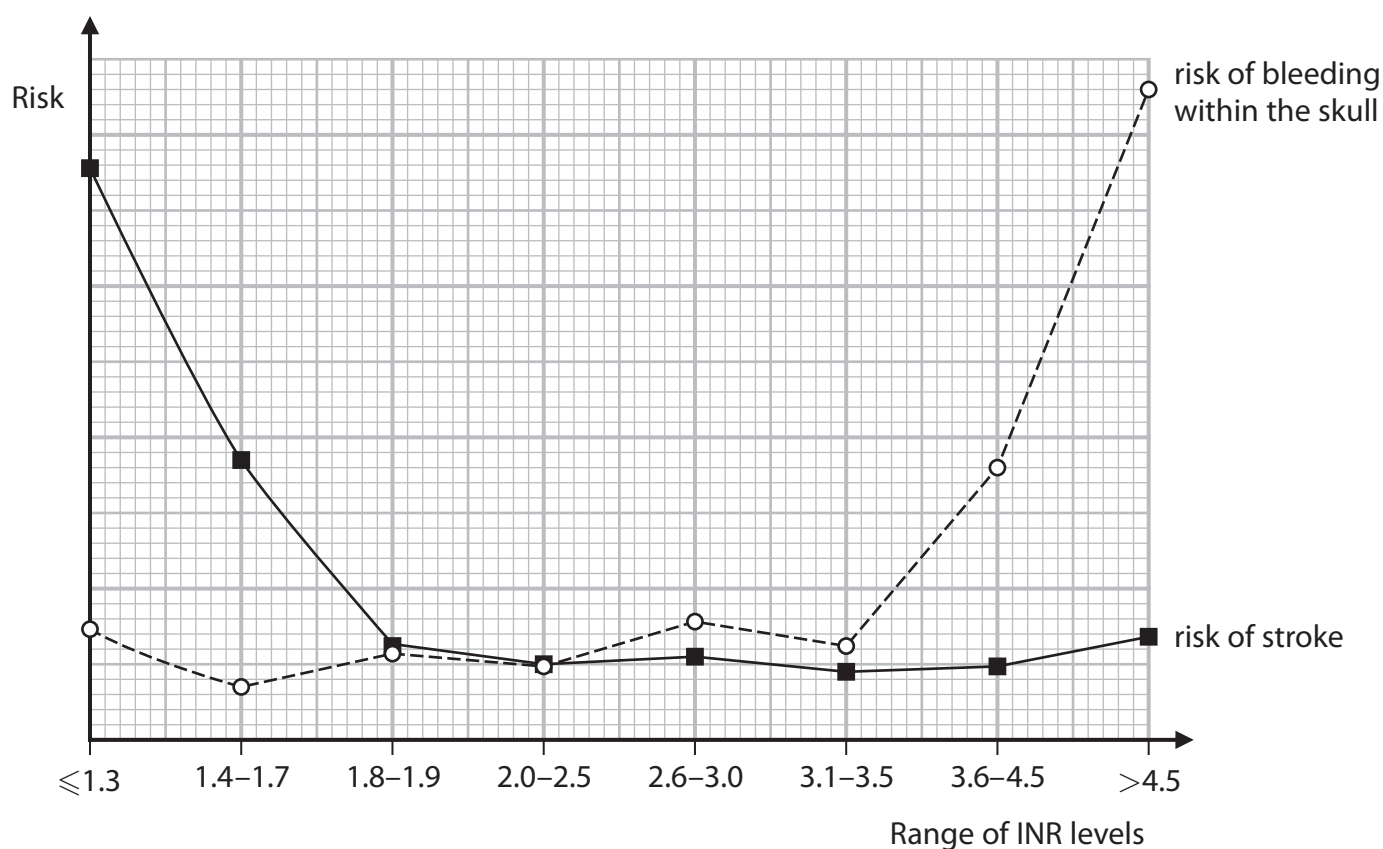
☐ **A** none

☐ **B** one

☐ **C** two

☐ **D** three

(b) The graph shows the relationship between the INR level and the risk of stroke and the risk of bleeding within the skull.



Deduce a safe range for the INR level.

Use the information in the graph to support your answer.

(3)

- (c) Explain how taking this anticoagulant can reduce the risk of cardiovascular disease (CVD).

(3)



- (d) Warfarin is given to patients who have had a heart attack, to prevent further heart attacks and other related health issues.

A study compared warfarin with the use of aspirin, over a four-year period following the first heart attack.

- (i) A group of 1216 were given warfarin; 83.3% of these people had no further health issues.

Calculate the number of people given warfarin who had no further health issues following their first heart attack.

(1)

Answer

- (ii) A second study compared this result with people given a combination of warfarin and aspirin.

Describe how this study should be carried out to produce a valid comparison.

(4)

(Total for Question 6 = 12 marks)

